MMA3001 Project Report

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Introduction - The Engineering Problem

When it comes to machining mechanical parts, efficiency is everything. CNC milling machines need to produce results fast, but that speed comes at the cost of wearing out the tools themselves. This becomes a problem when the tolerances demanded are tight, because worn tools lead to poorer surface quality and can cause larger vibrations in the machine itself (JLCCNC, 2025). The rate at which the tools wear out then becomes an important economic parameter, as increased wear rate requires more frequent tool changes and maintenance. On top of the increased downtime, the cost of the replacement bits can add up, leading to an overall increase in production cost.

The goal of this project was to develop a numerical model that could predict the flank wear on a tool over time given multiple different operating parameters and materials being cut, which machinists could then use to determine the optimal settings to balance speed and maintenance. This was done by analyzing a public NASA dataset by Goebel & Agogino (2022), researchers at NASA and UC Berkeley respectively, who tested flank wear using a Matsuura MC-510V milling machine. The data was recorded in multiple cases, for each case the researchers would start with a fresh tool and run it at certain settings until the tool was adequately worn, stopping every few minutes to remove the tool and record the wear. The researchers also recorded time series data while the machine was running such as vibration, acoustic emission, and the current drawn. Due to the limitations of the dataset provided, this project's scope only includes predicting wear on this specific machine at a specific spindle speed, and acts as a proof of concept of the method shown since collecting data on these milling machines is both expensive and time consuming (Li et al., 2025)

Inputs and Outputs

When using the neural network regression model I have created, the user can input the physical parameters on the machine they wish to use to estimate wear further in time.

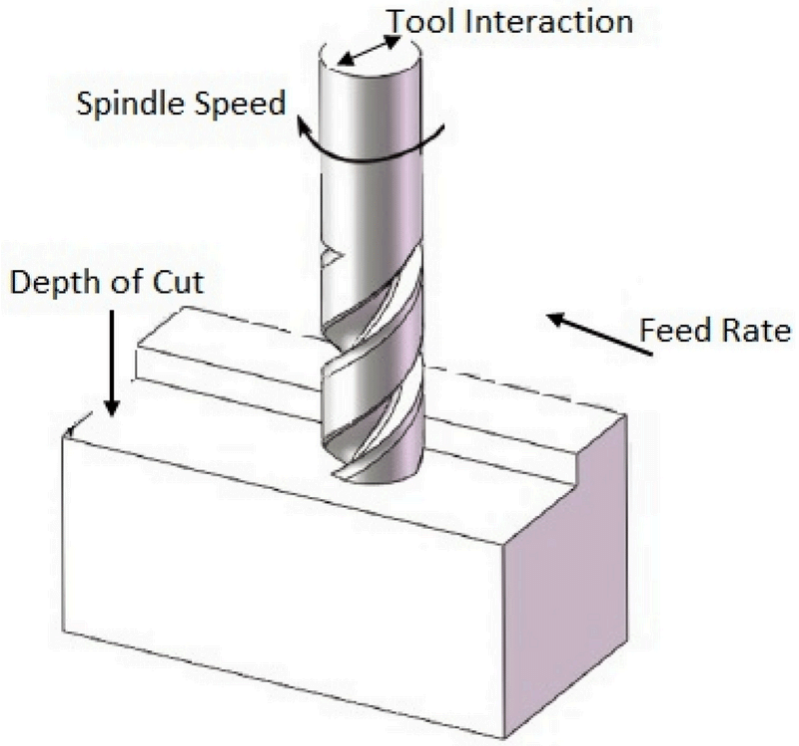


Figure 1: Diagram of physical parameters on a milling machine (Moayyedian et al., 2020)

As seen in figure 1, the solution accepts the machining parameters as inputs of Depth of Cut in mm, and feed rate in mm/min, for a constant spindle speed of 826 rev/min, which is what speed the data was collected at. It also must know the material being cut, which is a binary input system of 1 for cast iron or 2 for steel, as well as the time the machine has been running for in minutes. The output of the model is flank wear of the tool (VB), which is measured in mm and can be used to estimate when the tool will need to be replaced. Before training the model, a function was created to remove all missing entries, as well as all cases that had only had a single entry at [0,0] and were therefore trivial to include. When using the model, user inputs are rejected if they're not supported and the user will be prompted to enter the desired parameter again.

While experimenting with creating the model, I found that including the average spindle and average table vibration magnitudes from the time series entries actually resulted in a slightly lower validation RMSE. However, this is not data that machinists will know in advance, and therefore could not be used to predict wear in the future. I have included this version of the model as an extension at the end of the main notebook, as it could still be

useful for estimating current wear while running, given the proper sensors are installed on the machine.

Computational Solution

My solution to this problem was to train a neural network regression model using the Scikit-learn functions, and including a standard scaler in the training pipeline as taught in notebook 5.4 to ensure features are considered equally. Neural networks were chosen due to their ability to learn complex relationships between variables, especially when in this case, there are no existing physical laws or relationships between them. Goebel & Agogino (2022) stated that researchers have traditionally tried to model tool wear using mathematical formulas such as Taylor's tool wear formula, which relates tool life to cutting speed using experimentally derived constants. These methods have shown only moderate success due to the amount of factors at play, making machine learning well suited for this problem.

Importantly, we must keep in mind the limited scope of the data provided. Due to the expensive nature of milling, all data points used for training have the same spindle speed, machine type, and tool type, meaning the model may be poor at estimating wear for other machines or spindle speeds. There are also plenty of other factors in the machining process not included in the dataset that could impact the wear, such as other materials being cut and their densities, the amount or type of coolant used, the shape of the bit, the torque on the bit, etc. Judging by its performance here, a neural network would be great at incorporating these factors as well given the data, but currently the data has simply not been recorded.

Alternative Solutions

When evaluating the model's performance, it was compared against a linear regression model as a baseline.

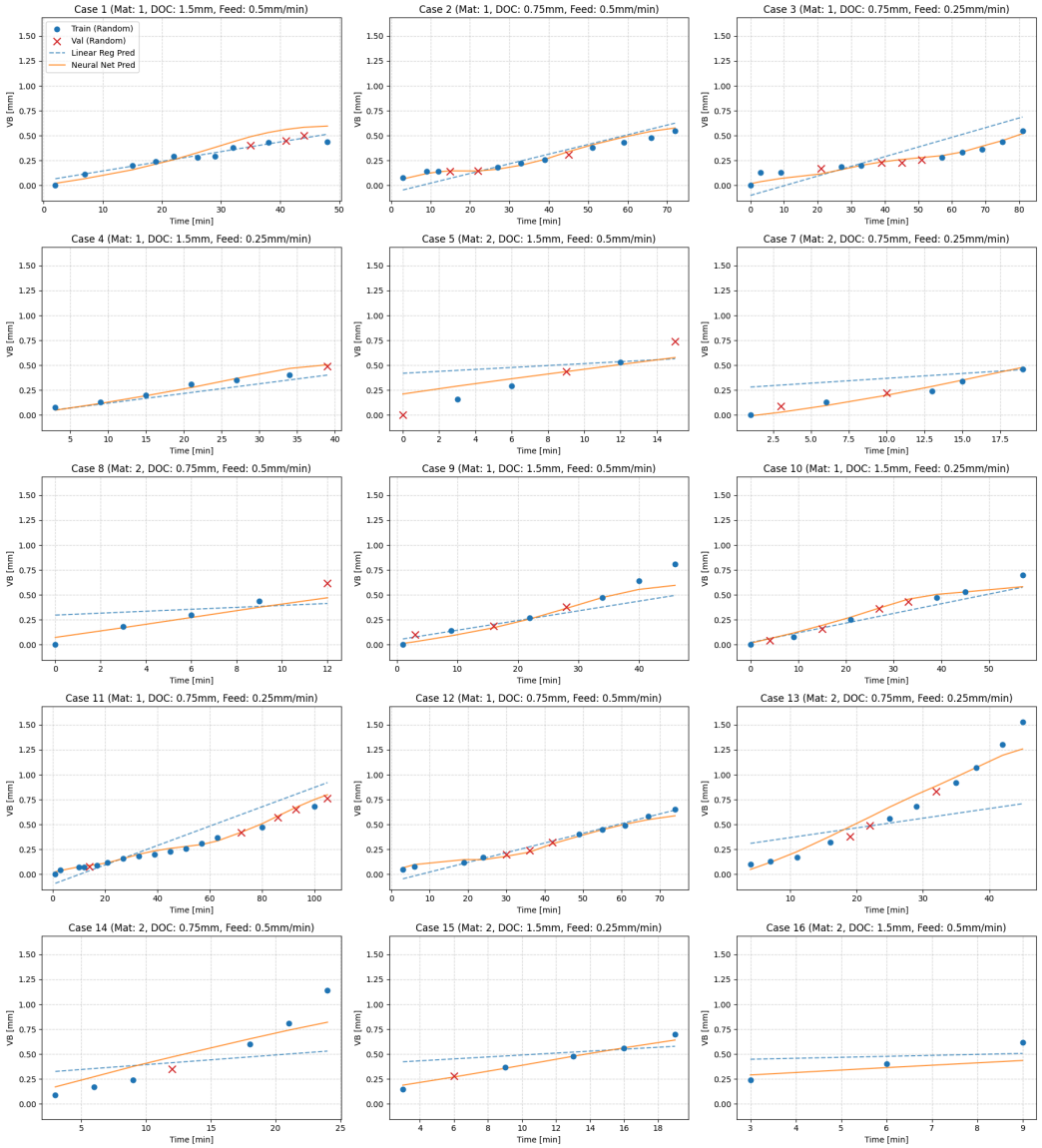


Figure 2: VB vs time for all cases, including Neural Network, linear regression models and training/validation split

It was found that the linear model was more poorly suited for the task, having a validation RMSE of 0.1278 mm, compared to the neural network with 0.0691 mm. The linear model also proved to be less flexible when given multiple features, giving a worse RMSE when given the time series data. When plotted against time, the wear rate tends to increase slightly over time (Figure 2), making linear regression a poor approximation. It's for this reason that other methods such as decision tree regression were not considered, the wear is best modelled as a smooth curve. When it comes to performance, while linear regression is

less intensive, this problem was not very computationally limited and the additional compute time of the neural network was trivial.

Validation

The performance of the model was validated by using a random validation/training split, where 75% of data entries were used for training, and the final 25% were used for validation (Figure 2). Initially, I attempted to use a chronological future data split with the final 25% in time being used for validation, but both models proved to be poor at extrapolation (Appendix 1) and I was advised by a TA to try a random split instead. Given the model's intended use, random split is suitable for predicting wear on the same machine under similar time spans. Given the unpredictabilities of the wear process, the uncertainty of the data, and that wear is typically measured in tenths of millimetres, an RMSE of 0.0691 mm is a fairly good result for unseen data. To ensure the result is reproducible for future use, the data was split using a set random seed.

Optimization and Performance

Due to the fairly modest amount of data points, being around 150 after filtering the NAN values and trivial entries, this was not a very computationally challenging solution. The neural network used two layers of 64 and 32 neurons respectively and a ReLU activation function, which was found through trial and error. Similarly, the model only used 1000 iterations in training, as I found it had converged to a final solution by then, and more iterations would be needlessly adding compute power.

Conclusion

In summary, I believe that this project has helped me learn the process of selecting, implementing, and refining numerical methods and machine learning to tackle a real engineering problem. The neural network model shows great promise in accurately modelling

and predicting CNC mill tool wear, due to its unique ability to adapt to the immense amount of factors, much more so than any traditional mathematical technique. My biggest takeaway from this project is the sheer importance of having a large, reliable dataset. In its current state, the model can reliably predict wear under quite a few different settings, but is still limited by the specific machine, spindle speed, and material selection of which it was trained on. If machinists were to collect more data for their respective machining conditions, this model would prove very useful at improving efficiency and reducing costs.

AI Use and Reflection

Throughout this project, ChatGPT (chatgpt.com) and Google Colab's built-in Gemini assistant (gemini.google.com) were used to assist with both research and development of the solution. ChatGPT was primarily used for helping with theory and concepts, such as providing informational YouTube videos on how CNC mills work, or helping search for suitable public datasets that fit the project brief. It was also used to help explain the process of using Github, as I had never used it before and found it quite confusing. When using it for research, I made sure to use its output as a starting point rather than the ground truth, and compared what it said with trusted websites and course material. Gemini was mainly used for assisting to write the code in the notebooks themselves to speed up the process. All code outputs from the AI were thoroughly checked by myself using my professional engineering judgement, ensuring I understood what it was doing and testing to ensure the results were acceptable. All important decisions, such as which models to try, which validation splits to use, etc. were made by myself. All documentation was written by myself with no assistance.

I found that the AI was not very good at deciphering the physical meaning of what it was being asked to do, and required very specific instructions to avoid misinterpreting the data. It also struggled with problems specific with the dataset, such as how to handle the time series data being recorded differently from the rest. In these cases, I found consulting Dr. Atkinson or a TA to be much more helpful. Overall, the use of gen AI was quite helpful assisting with the more monotonous aspects such as the coding, and helped improve my understanding of how the concepts we've been learning can be applied in practice.

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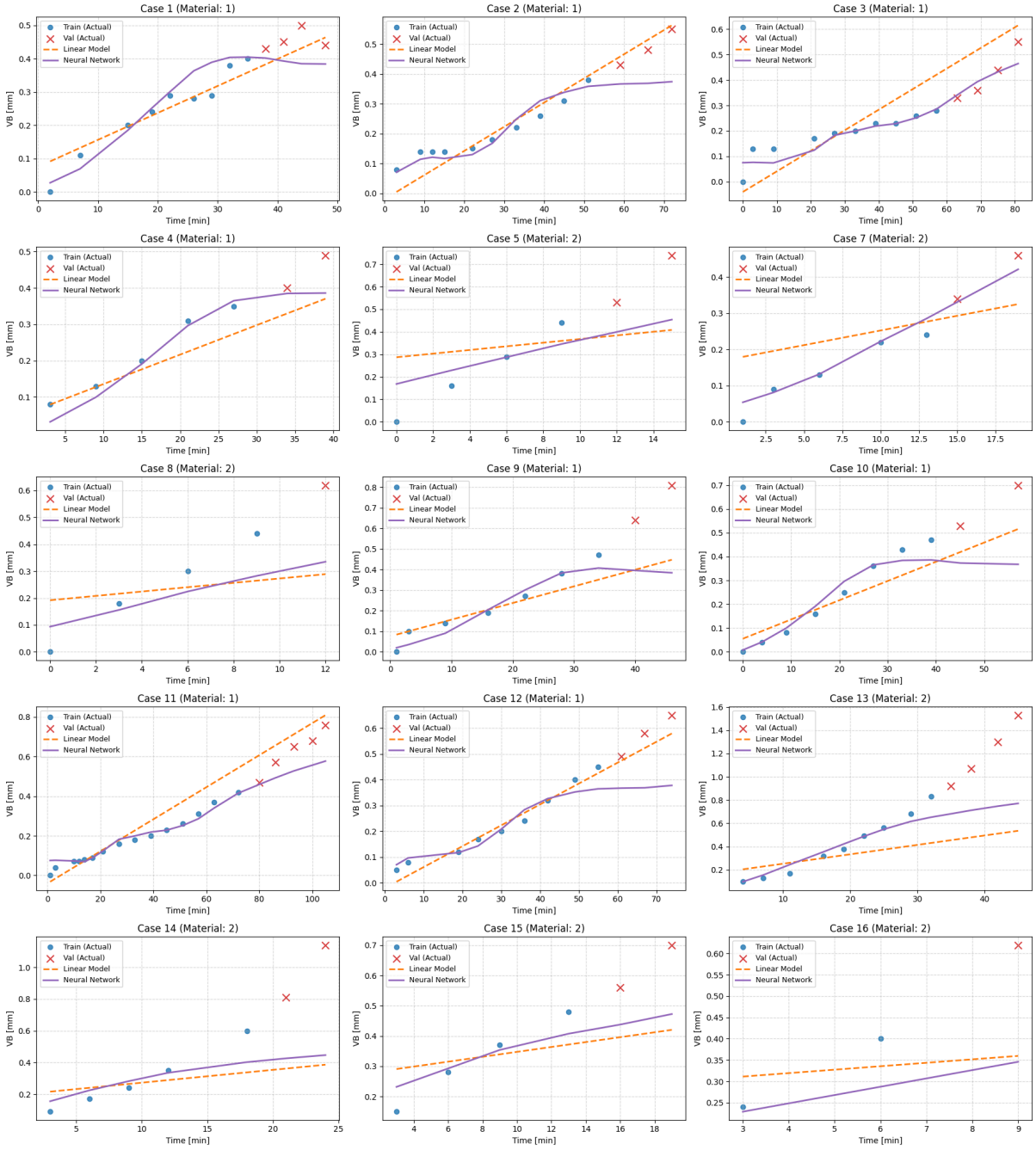
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Appendix



A1: Model results when using a future training split (Not used in final product)